19、Dcm $22 $2E服务

卖家'漩涡遨游aa' 专属资料，非在漩涡遨游aa 购买资料，不提供答疑(一次购买仅提供有限次数10次答疑)

ISO14229-1规定了26个诊断服务细节，也就是UDS诊断报文的细节。它只规定了各个服务每个字节的含义，它不关心底层到底是怎么传输的。

ISO15765-2规定了基于CAN总线进行UDS报文传输的细节（包括四种帧）。是在CAN总线传输的情况下， 具体规定如何传输UDS诊断的请求和响应报文的。四种帧就是定义在这个标准中。

UDS诊断报文分为“单帧传输”，“多帧传输”两种。

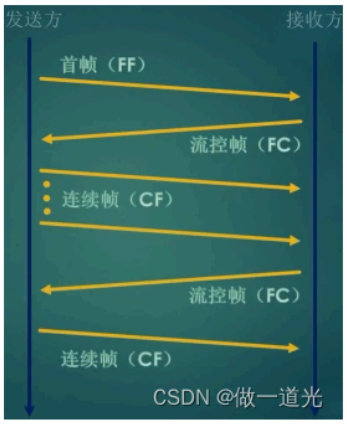
单帧传输

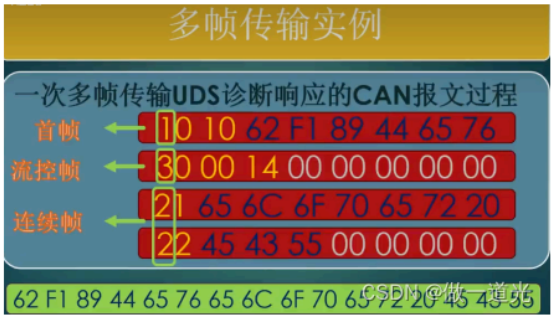
一个UDS的请求或者响应的发送方，发出一个单帧（SF）即可，接收方接收到就可以了。因为单帧的时候，这一帧CAN报文是能够装得下所有的uds报文。

一个单帧UDS诊断，请求或响应CAN报文，第一个字节第一位为0.

多帧传输

一个数据没办法用一帧CAN报文发送完，就需要用多帧（FF）传输。流控帧（FC）负责根据接收方和发送方的实际需要，来实时的调节下次发送连续帧（CF）时如何发送。





如下图所示，是一个接收方发送给发送方的流控帧（FC）示例。

一帧CAN报文中有8个字节，流控帧的控制信息一定是3个字节，还有5个字节用00填充，称为“补白”。

1、N_PDU类型

N_PDU类型	N_PCI			
	Byte 1		Byte 2	Byte 3
	Bit 7~4	Bit 3~0		
单帧（SF）	N_PCIType=0	SF_DL<=7[1]	N_Data	
第一帧（FF）	N_PCIType=1	7<FF_DL<=4095		N_Data
连续帧（CF）	N_PCIType=2	SN（0~F循环计数）	N_Data	
流控帧（FC）	N_PCIType=3	FS	BS	STmin

CSDN @Bhind

❖ FS含义

➢0：继续发送。让发送方继续发送接下来的连续帧，表示接收方已经准备好了接收最大为BS数量的连续帧。
➢1：等待。发送方等待下一帧流控帧并重置自己的计时。一般用于接收方没有处理完上一次接收到的连续帧。
➢2：过载。发送方打算发送的数据长度超过了接收方的储存能力。

❖ BS含义：1~FF表示发送方在发送BS数值的连续帧之后，需要等待接收方的流控帧。0表示不需要任何流控帧，直接发送全部数据。

❖ STmin含义：表示发送方发送的两个连续帧之间，**最小**需要的时间间隔。00~7F单位为ms；F1~F9表示0.1~0.9ms；0表示按照发送方最快的速度发送。

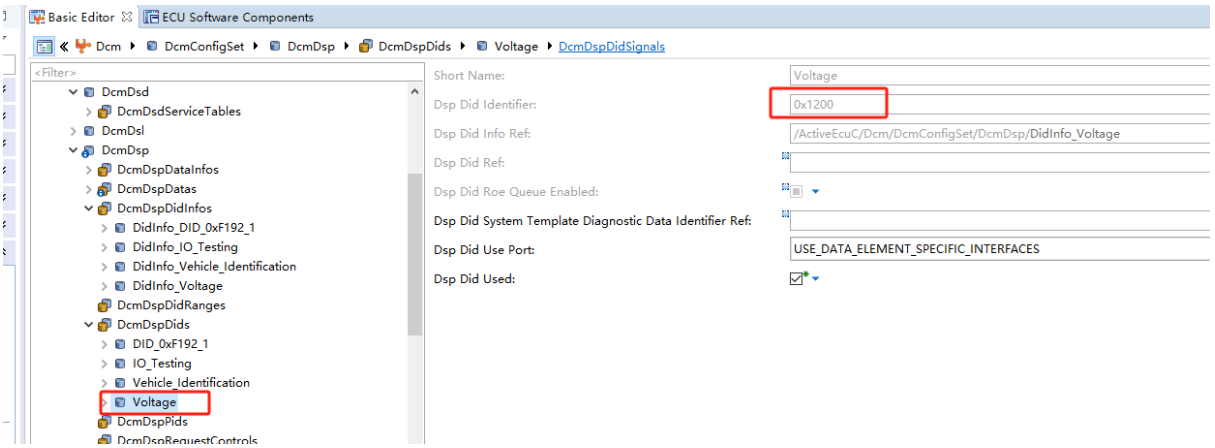
CSDN @Bhind

STmin：体现了接收方的处理能力，处理CAN接收的速度
BS：体现了接收方的接收能力

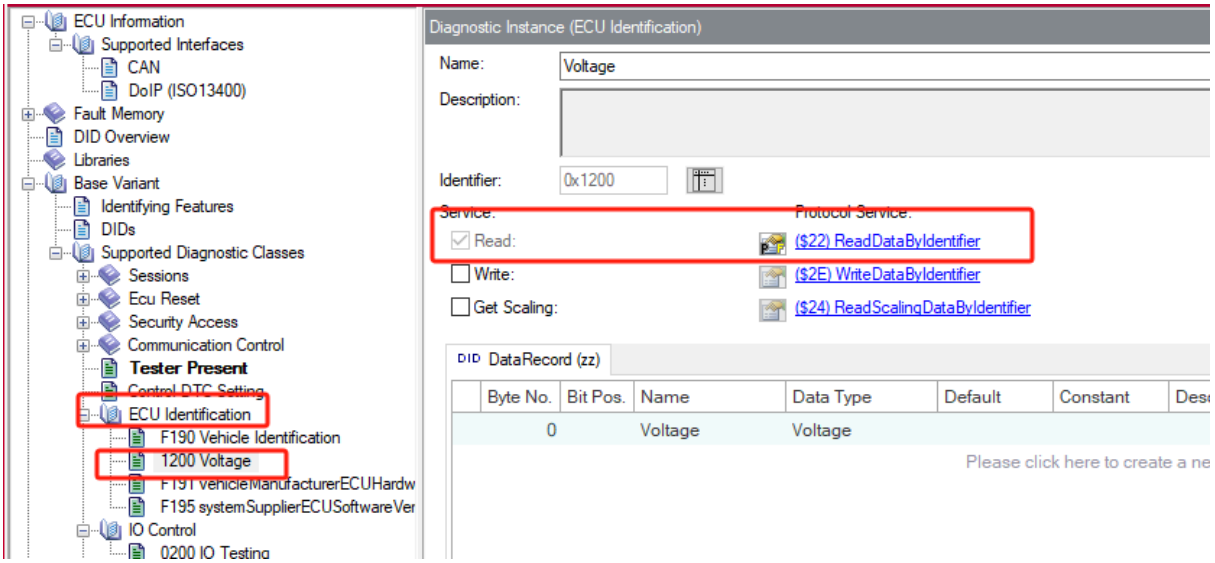
一、22服务读DID

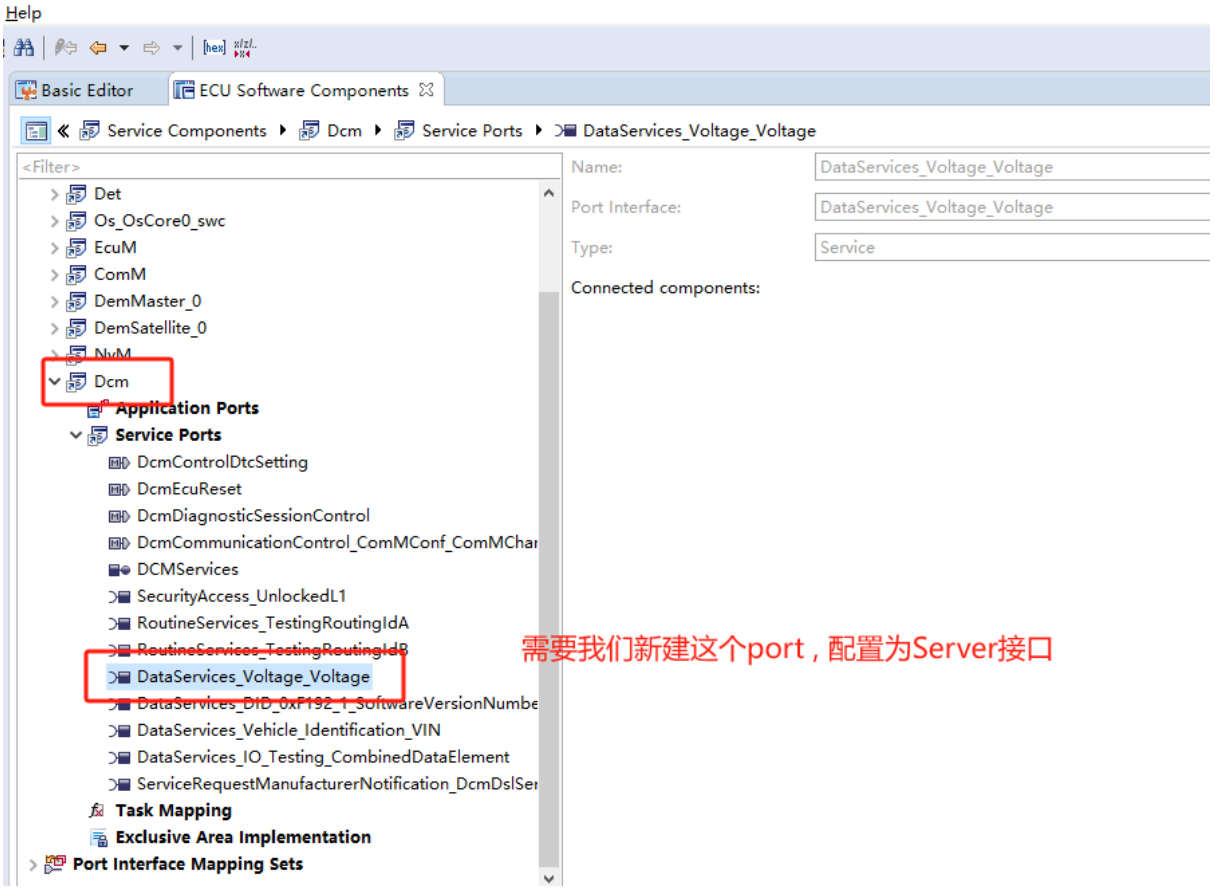
1.1、developer添加DID 0x1200服务接口

以DID 0x1200为例

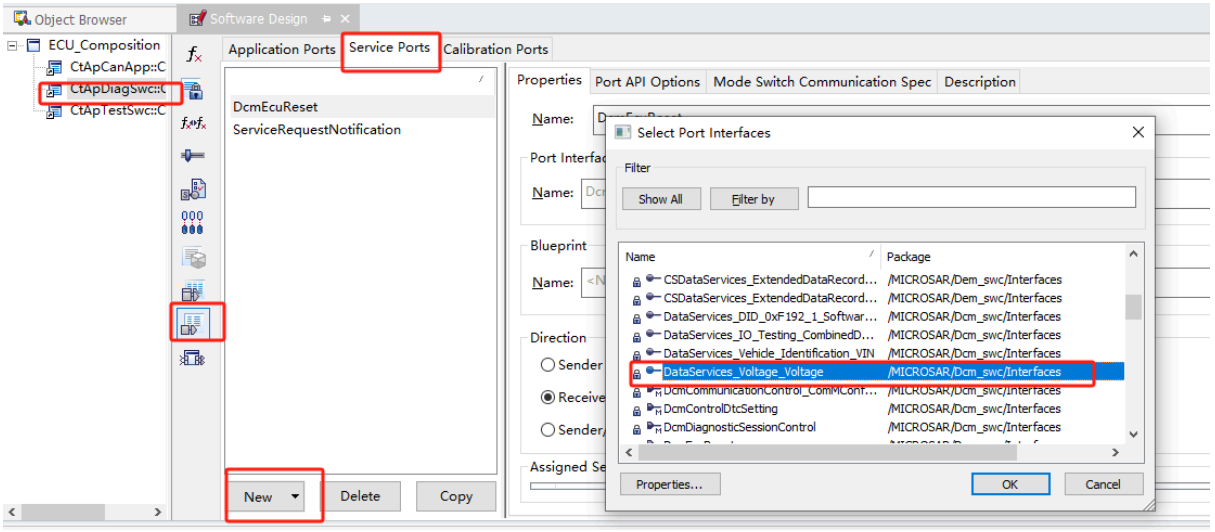


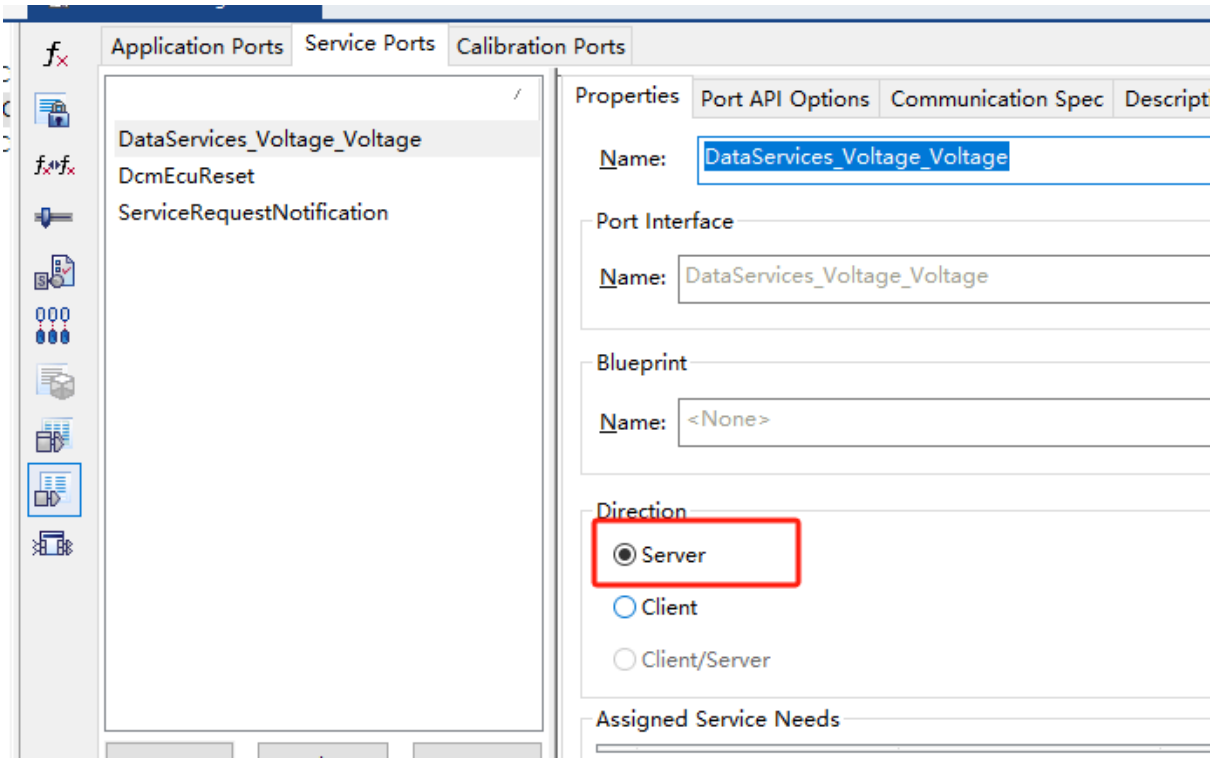
通过看cdd文件，这个did两个字节，只有22， 没有2E



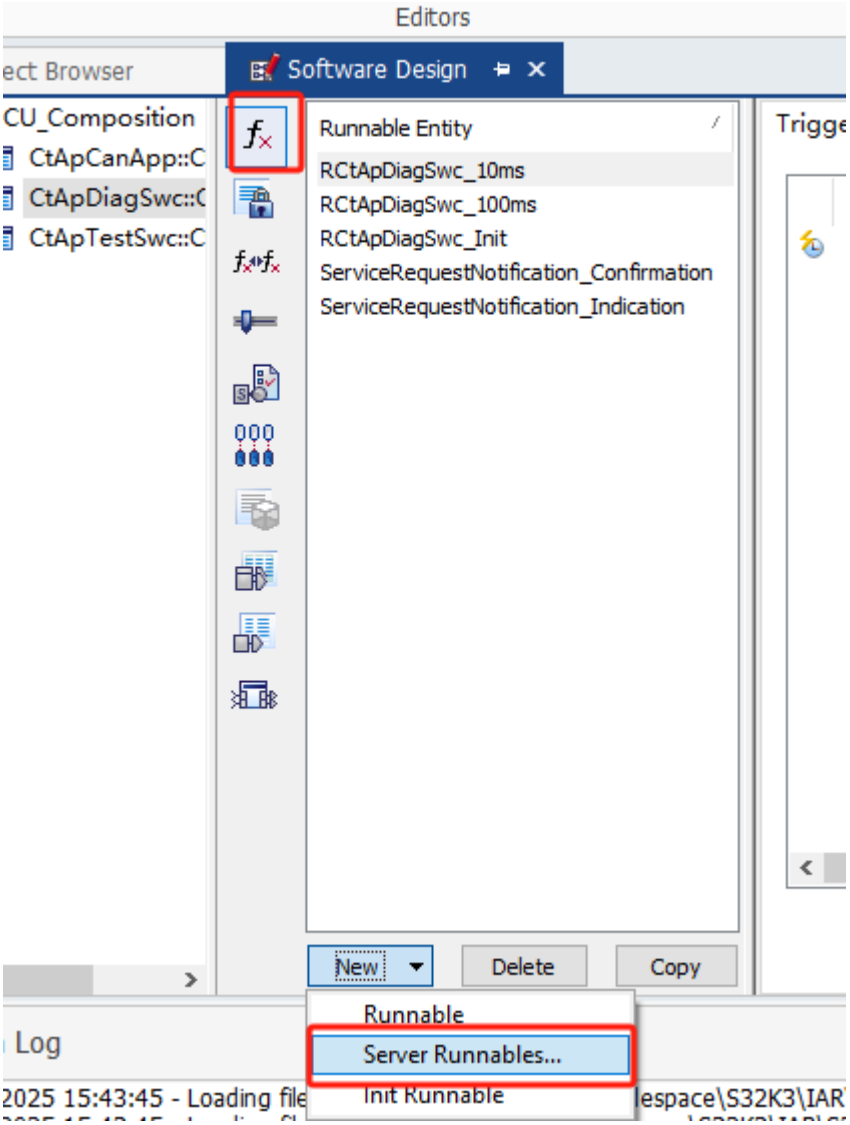


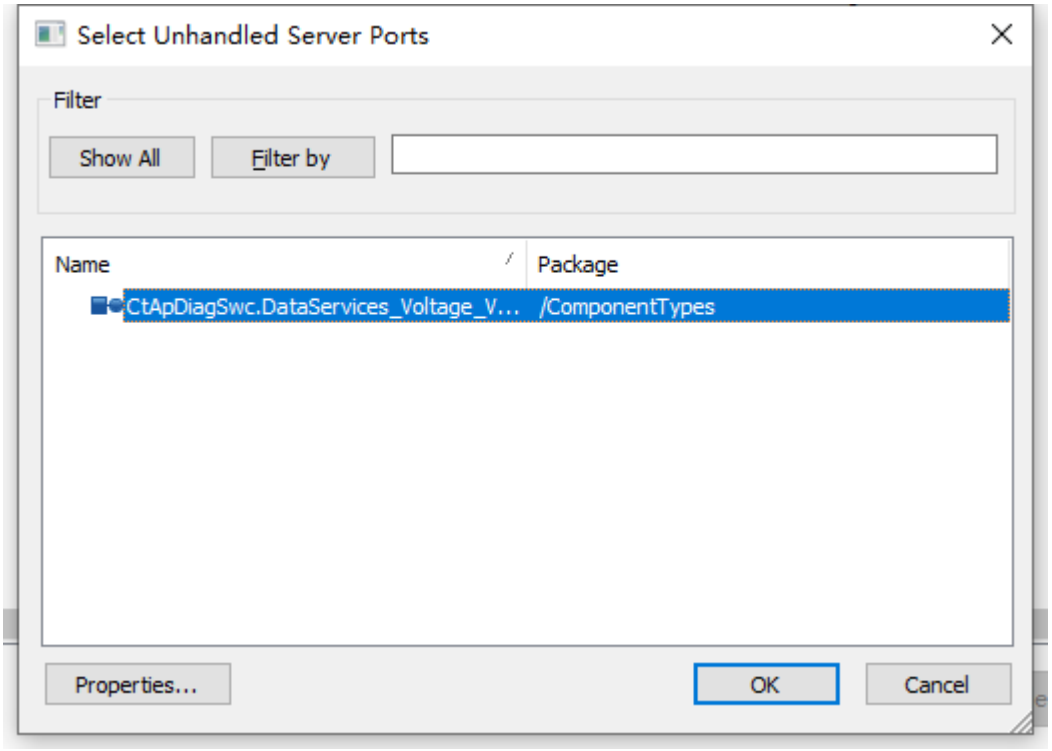
打开developer, 在CtApDiagSwc中新建DataServices_Voltage_Voltage 接口并配置为Server

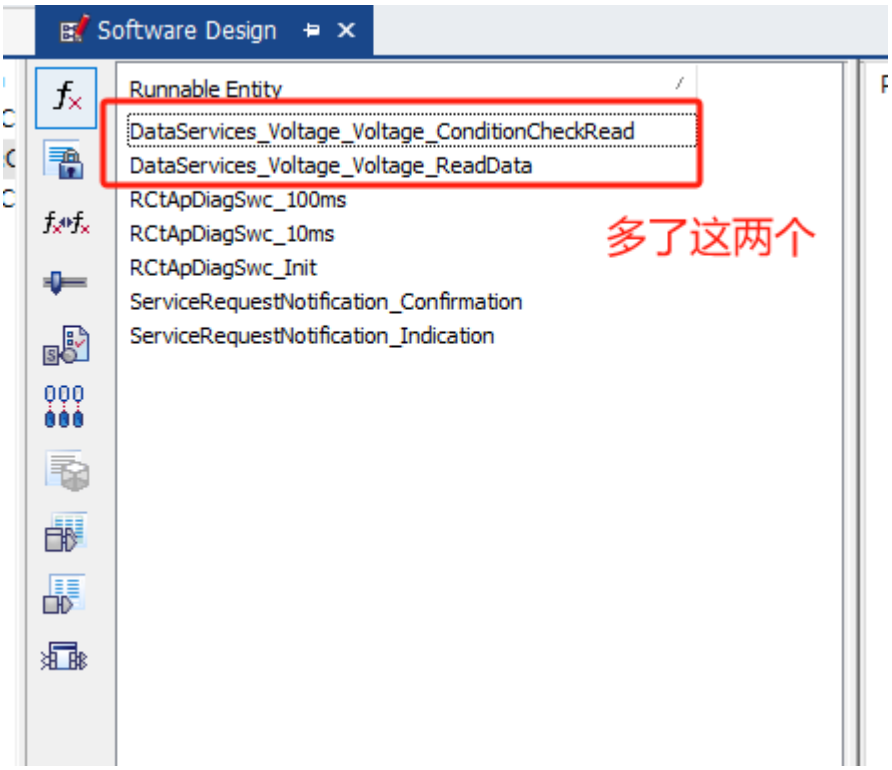

Object Browser
Software Design
ECU_Composition
CtApCanApp::C
CtApDiagSwc::C
CtApTestSwc::C
Application Ports
Service Ports
Calibration Ports
DcmEcuReset
ServiceRequestNotification
New
Delete
Copy
Properties
Port API Options
Mode Switch Communication Spec
Description
Name:
Port Interfac
Blueprint
Direction
Sender
Receive
Sender/
Assigned Se
Select Port Interfaces
Filter
Show All
Filter by
Name
Package
CSDataServices_ExtendedDataRecord...
/MICROSAR/Dem_swc/Interfaces
CSDataServices_ExtendedDataRecord...
/MICROSAR/Dem_swc/Interfaces
DataServices_DID_0xF192_1_Softwar...
/MICROSAR/Dcm_swc/Interfaces
DataServices_IO_Testing_CombinedD...
/MICROSAR/Dcm_swc/Interfaces
DataServices_Vehicle_Identification_VIN
/MICROSAR/Dcm_swc/Interfaces
DataServices_Voltage_Voltage
/MICROSAR/Dcm_swc/Interfaces
DcmCommunicationControl_CommConf...
/MICROSAR/Dcm_swc/Interfaces
DcmControlDtcSetting
/MICROSAR/Dcm_swc/Interfaces
DcmDiagnosticSessionControl
/MICROSAR/Dcm_swc/Interfaces
Properties...
OK
Cancel


Application Ports
Service Ports
Calibration Ports
DataServices_Voltage_Voltage
DcmEcuReset
ServiceRequestNotification
Properties
Port API Options
Communication Spec
Descript
Name:
DataServices_Voltage_Voltage
Port Interface
Name:
DataServices_Voltage_Voltage
Blueprint
Name:
<None>
Direction
Server
Client
Client/Server
Assigned Service Needs

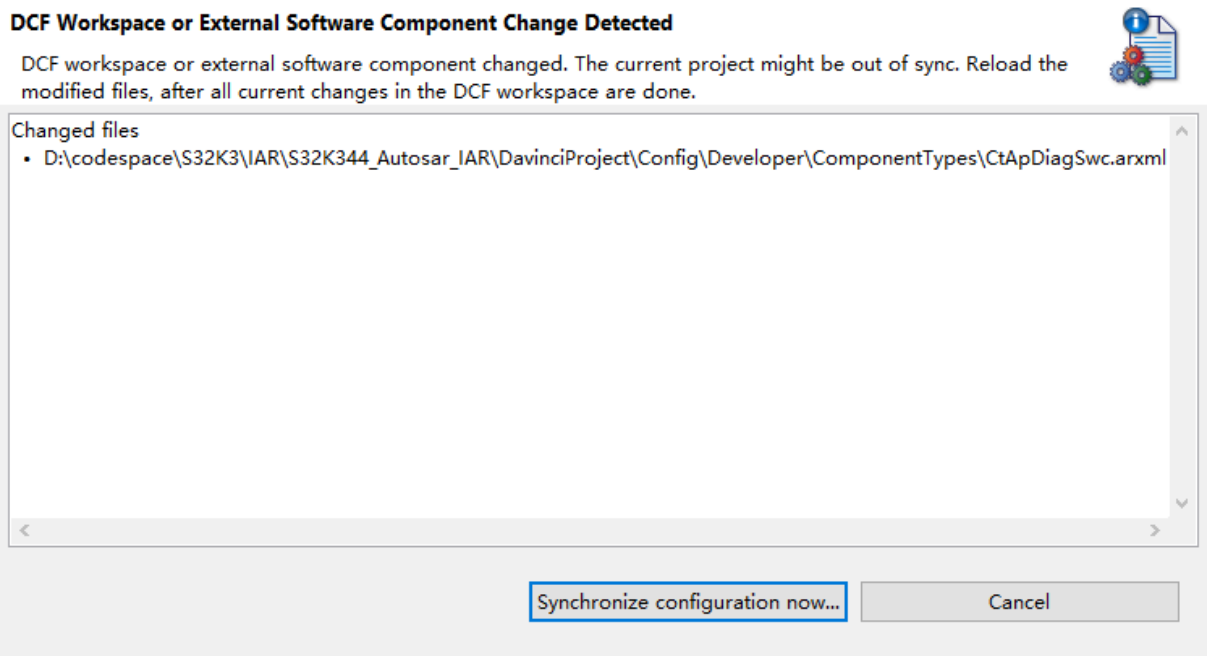
添加服务函数


Editors
ect Browser
Software Design
CU_Composition
CtApCanApp::C
CtApDiagSwc::C
CtApTestSwc::C
Runnable Entity
RCtApDiagSwc_10ms
RCtApDiagSwc_100ms
RCtApDiagSwc_Init
ServiceRequestNotification_Confirmation
ServiceRequestNotification_Indication
Trigge
New
Delete
Copy
Runnable
Server Runnables...
Init Runnable
Log
2025 15:43:45 - Loading file
lespace\S32K3\IAR

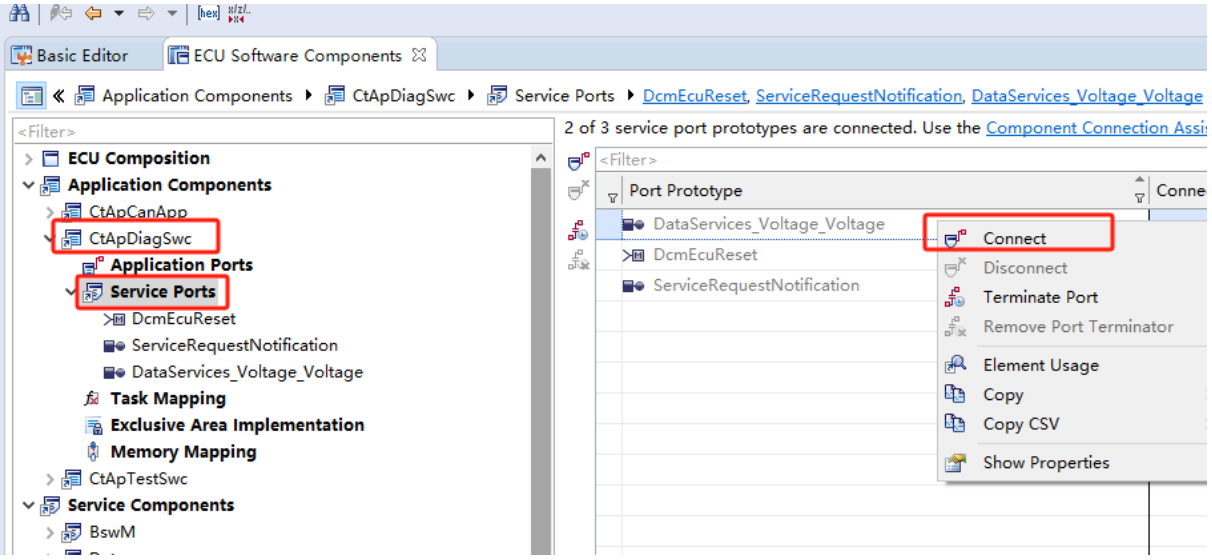

Select Unhandled Server Ports
Filter
Show All
Filter by
Name
Package
CtApDiagSwc.DataServices_Voltage_V...
/ComponentTypes
Properties...
OK
Cancel

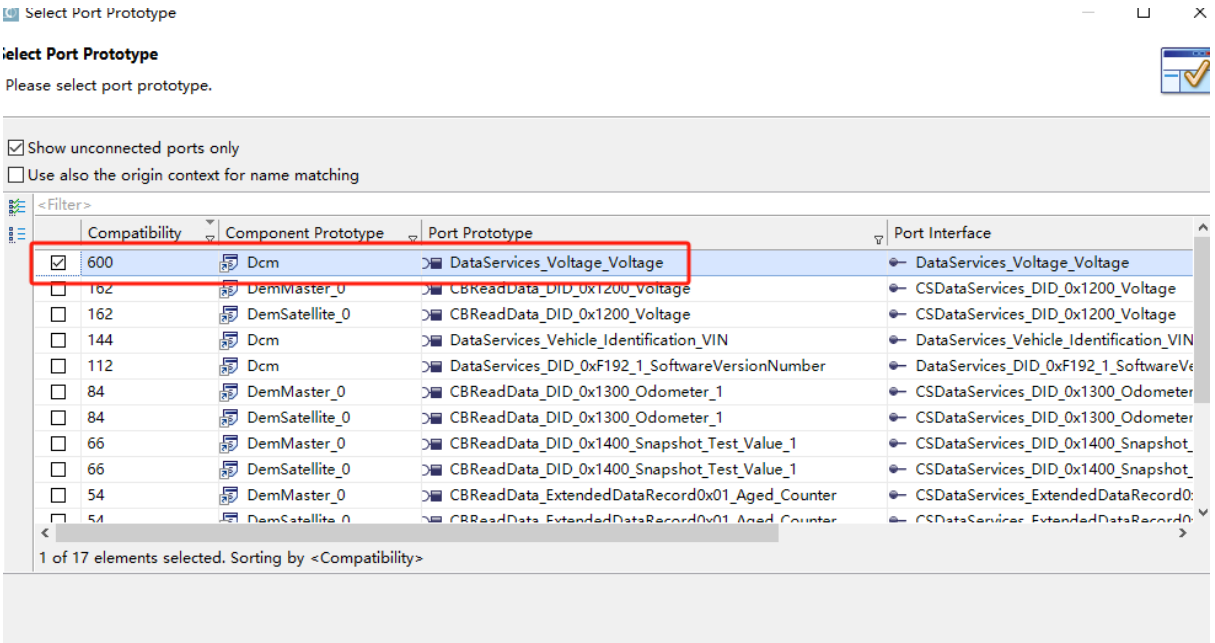

Software Design
Runnable Entity
DataServices_Voltage_Voltage_ConditionCheckRead
DataServices_Voltage_Voltage_ReadData
RCtApDiagSwc_100ms
RCtApDiagSwc_10ms
RCtApDiagSwc_Init
ServiceRequestNotification_Confirmation
ServiceRequestNotification_Indication
多了这两个

保存，然后去cfg 同步



1.2、cfg 连接Dcm DID0x1200的 client接口



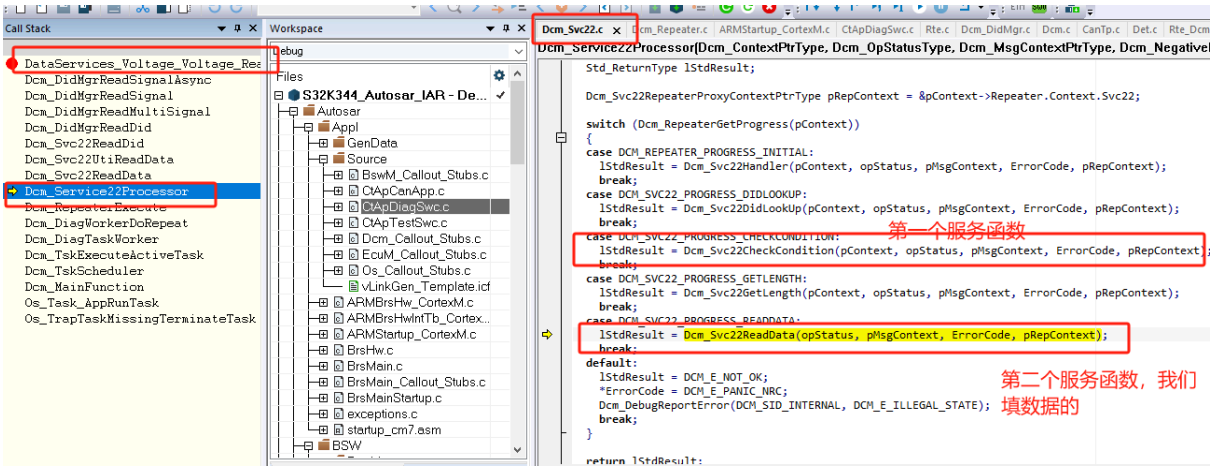


然后生成Rte/Os，再生成Swc代码，然后CtApDiagSwc.c 中就会多了两个服务接口

1.3、添加读DID内容的值

我们添加ReadData里面的实现内容

```
FUNC(Std_ReturnType, CtApDiagSwc_CODE) DataServices_Voltage_Voltage_ReadData(Dcm_OpStatusType OpStatus, P
{
/**********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          << Start of runnable implementation >>            DO NOT CHANGE
 * Symbol: DataServices_Voltage_Voltage_ReadData (returns application error)
 **********************************************************************************************************

  Data[0] = 0x45;
  Data[1] = 0x67;
  return RTE_E_OK;

/**********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          << End of runnable implementation >>              DO NOT CHANGE
 **********************************************************************************************************
}
```



回复正响应62，数据里面也包含了我们填写的数据

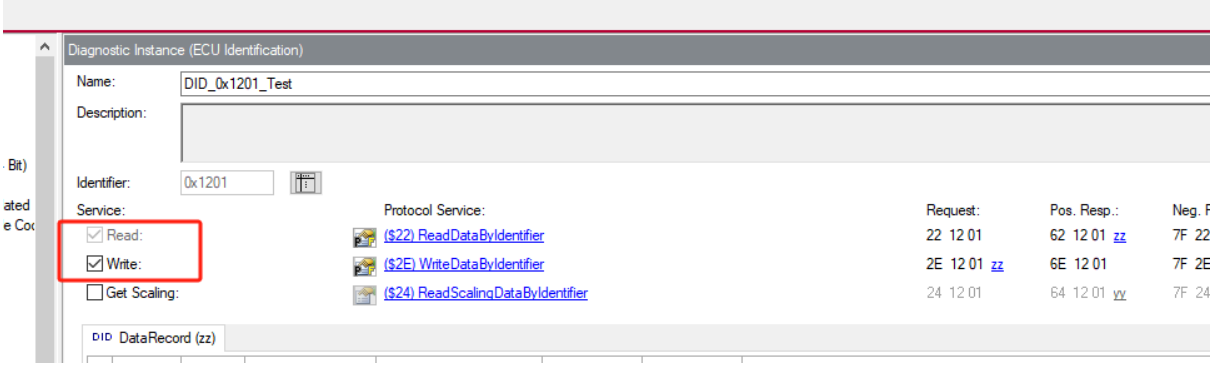
700	0	数据帧	Tx	8	03 22 12 00 00 00 00 00
708	0	数据帧	Rx	8	05 62 12 00 45 67 41 00

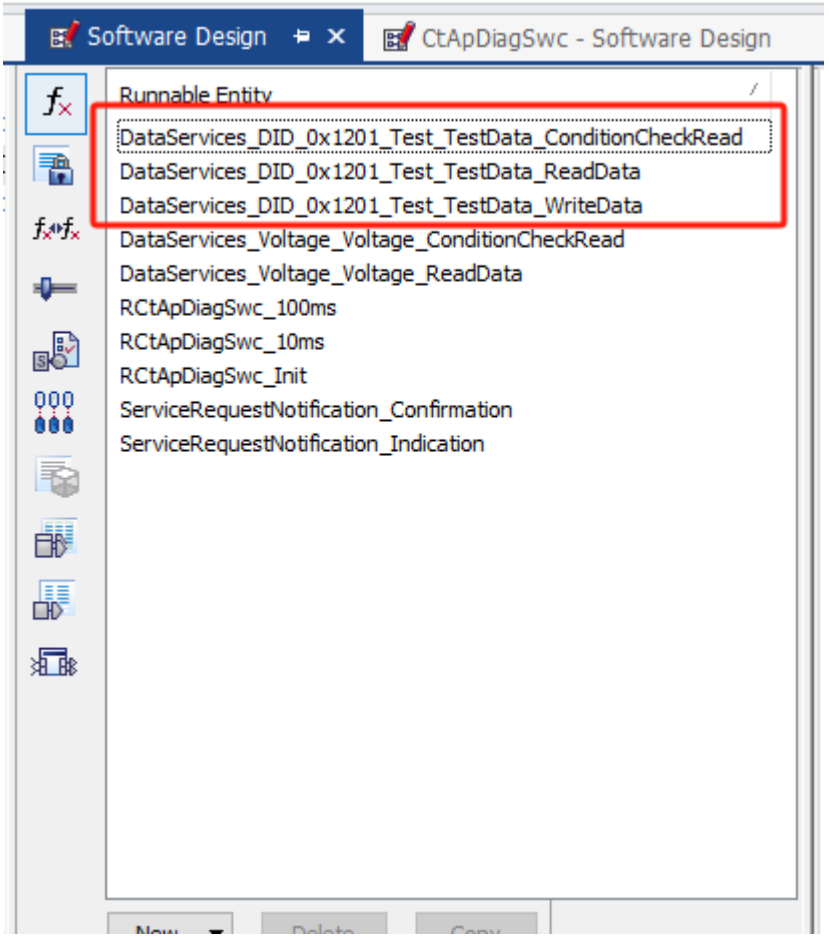
二、2E服务写DID

2.1、DataServices_DID_0x1201_Test_TestData添加port

按照1.1的步骤在CtApDiagSwc 添加DataServices_DID_0x1201_Test_TestData的port 并连接Dcm的client port， 这个port是一个字节的

有读写





2.2、代码添加读写接口

```
#include  Mcu.h
uint8 Did1201Data = 0x11;
/*************************************
```

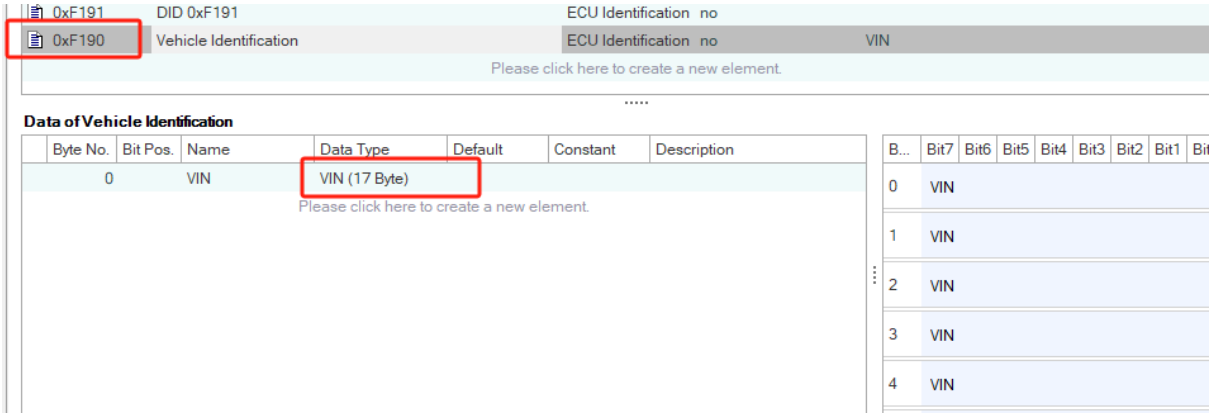
```
FUNC(Std_ReturnType, CtApDiagSwc_CODE) DataServices_DID_0x1201_Test_TestData_ReadData(Dcm_OpStatusType OpStatus, P2VAR(uint8,
{
/*****************************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          << Start of runnable implementation >>            DO NOT CHANGE THIS COMMENT!
 * Symbol: DataServices_DID_0x1201_Test_TestData_ReadData (returns application error)
 *****************************************************************************************************************/
  Data[0] = Did1201Data;
  return RTE_E_OK;

/*****************************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          << End of runnable implementation >>              DO NOT CHANGE THIS COMMENT!
 *****************************************************************************************************************/
}
```

```
FUNC(Std_ReturnType, CtApDiagSwc_CODE) DataServices_DID_0x1201_Test_TestData_WriteData(P2CONST(uint8, AUTOMATIC, RTE_CTAPDIAGSWC_APPL_DATA) Data, Dcm_OpStatusType
{
/*****************************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          << Start of runnable implementation >>            DO NOT CHANGE THIS COMMENT!
 * Symbol: DataServices_DID_0x1201_Test_TestData_WriteData (returns application error)
 *****************************************************************************************************************/
  Did1201Data = Data[0];
  return RTE_E_OK;

/*****************************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          << End of runnable implementation >>              DO NOT CHANGE THIS COMMENT!
 *****************************************************************************************************************/
}
```

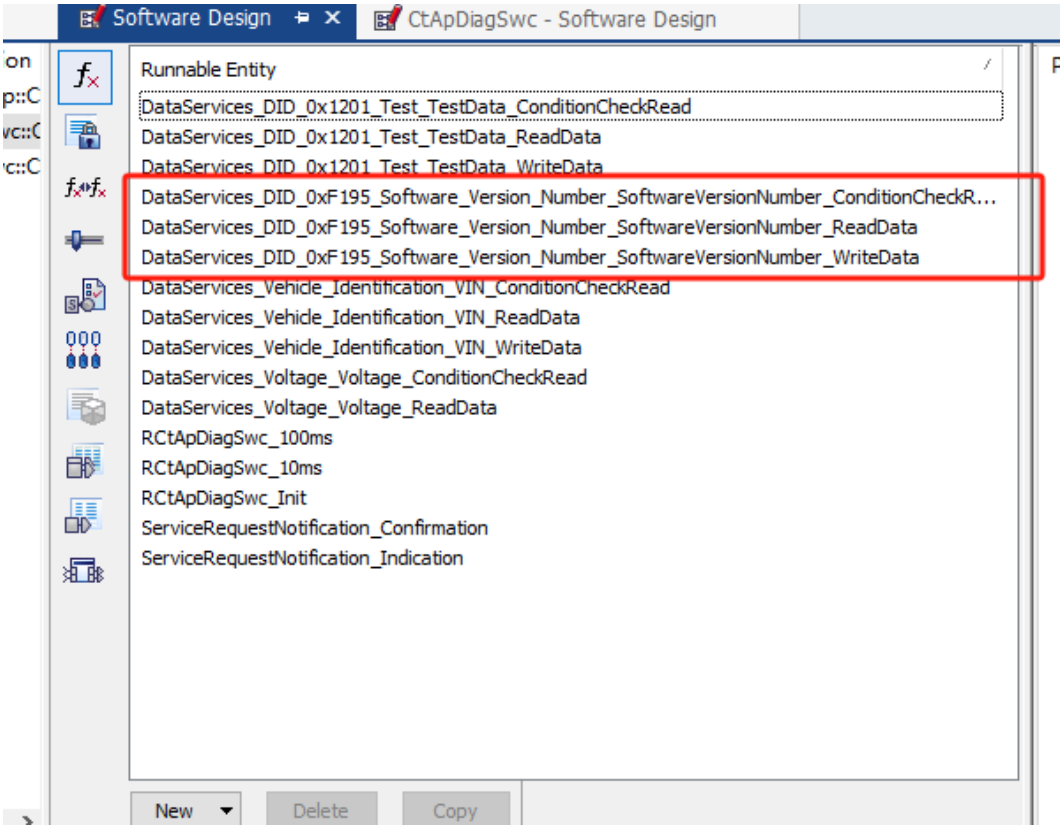
测试结果

三、$22 $2E服务的多字节读写

上面一、二都是一两个字节的读写，比较简单，交互一收一发即可，当DID是大于8字节是需要多帧的方式，下面以SoftwareVersionNumber的DID 为例，该DID是8字节



3.1、添加port, 同上面1.1步骤



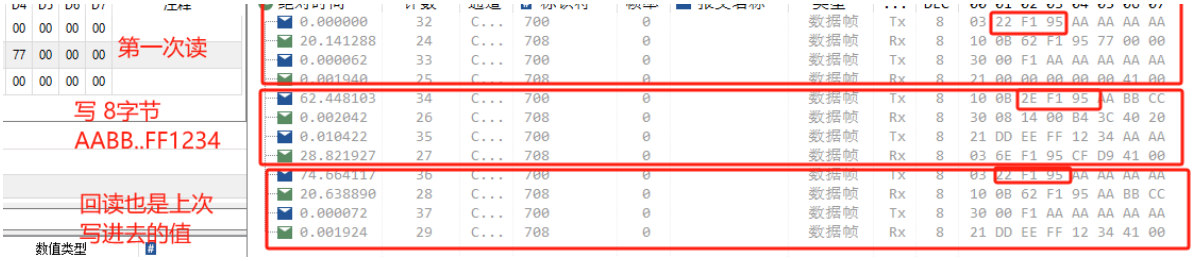

```
/*****************************************
 * DO NOT CHANGE THIS COMMENT!          << Star
 *****************************************
#include <string.h>
#include "Mcu.h"
uint8 Did1201Data = 0x11;
uint8 aDidF190Vin[17] = {0x99};
uint8 aDidF195SwVer[8] = {0x77};
/*****************************************
```

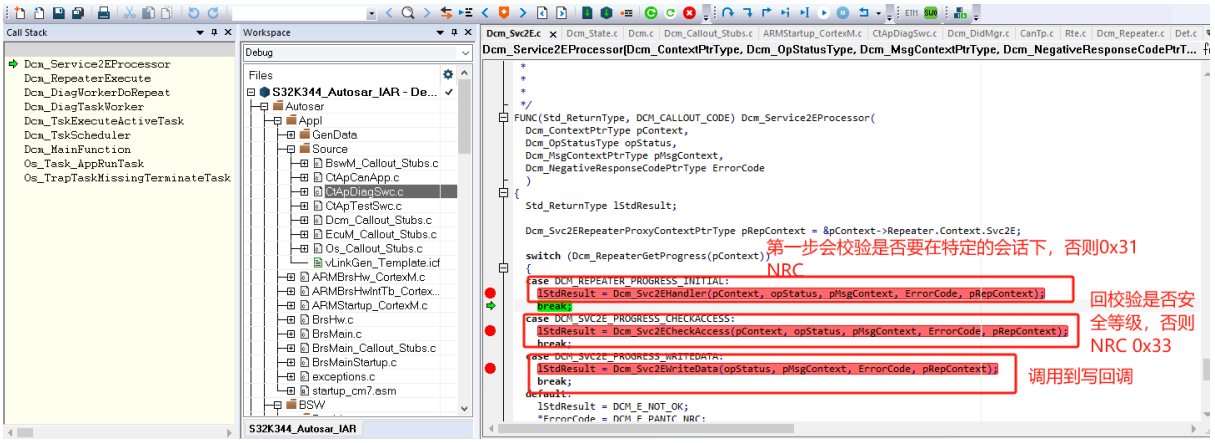
```
FUNC(Std_ReturnType, CtApDiagSwc_CODE) DataServices_DID_0xF195_Software_Version_Number_SoftwareVersionNumber_ReadData(
{
/**********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          << Start of runnable implementation >>            DO NOT CHANGE THIS COMMENT
 * Symbol: DataServices_DID_0xF195_Software_Version_Number_SoftwareVersionNumber_ReadData (returns application error)
 **********************************************************************************************************
  memcpy(Data, aDidF195SwVer, 8);
  return RTE_E_OK;

/**********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          << End of runnable implementation >>              DO NOT CHANGE THIS COMMENT
 **********************************************************************************************************
}
```

```
FUNC(Std_ReturnType, CtApDiagSwc_CODE) DataServices_DID_0xF195_Software_Version_Number_SoftwareVersionNumber_WriteData
{
/**********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          << Start of runnable implementation >>            DO NOT CHANGE THIS COMMENT
 * Symbol: DataServices_DID_0xF195_Software_Version_Number_SoftwareVersionNumber_WriteData (returns application error)
 **********************************************************************************************************
  memcpy(aDidF195SwVer, Data, 8);
  return RTE_E_OK;

/**********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          << End of runnable implementation >>              DO NOT CHANGE THIS COMMENT
 **********************************************************************************************************
}
```

3.2、验证多字节的读写

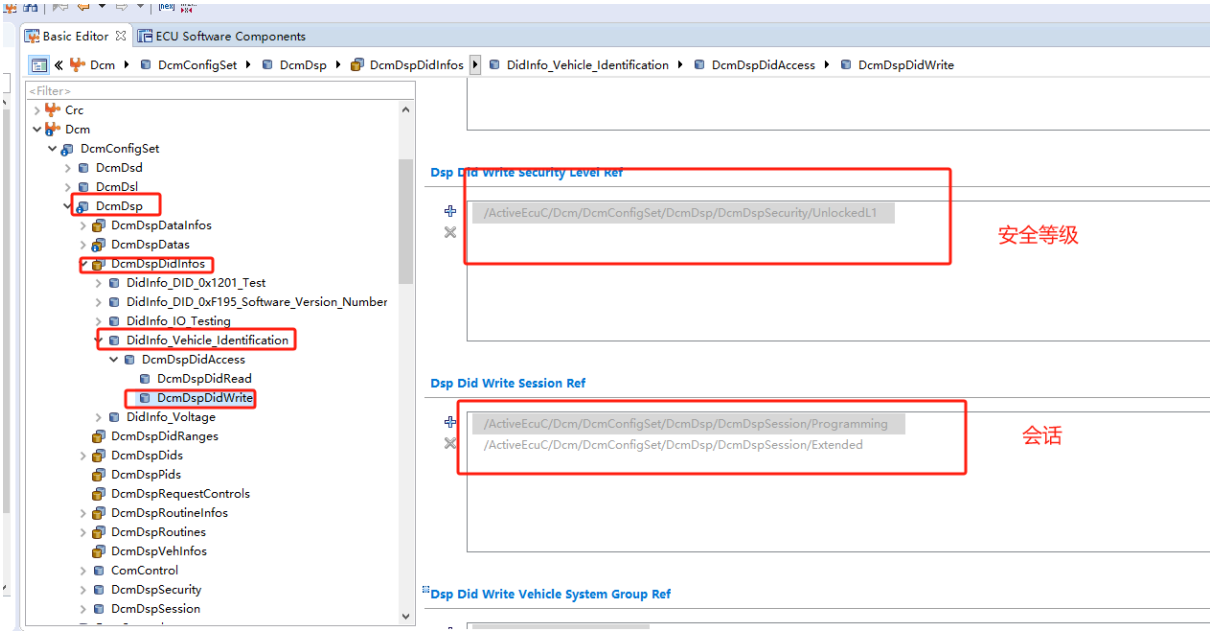
读写 22 F195





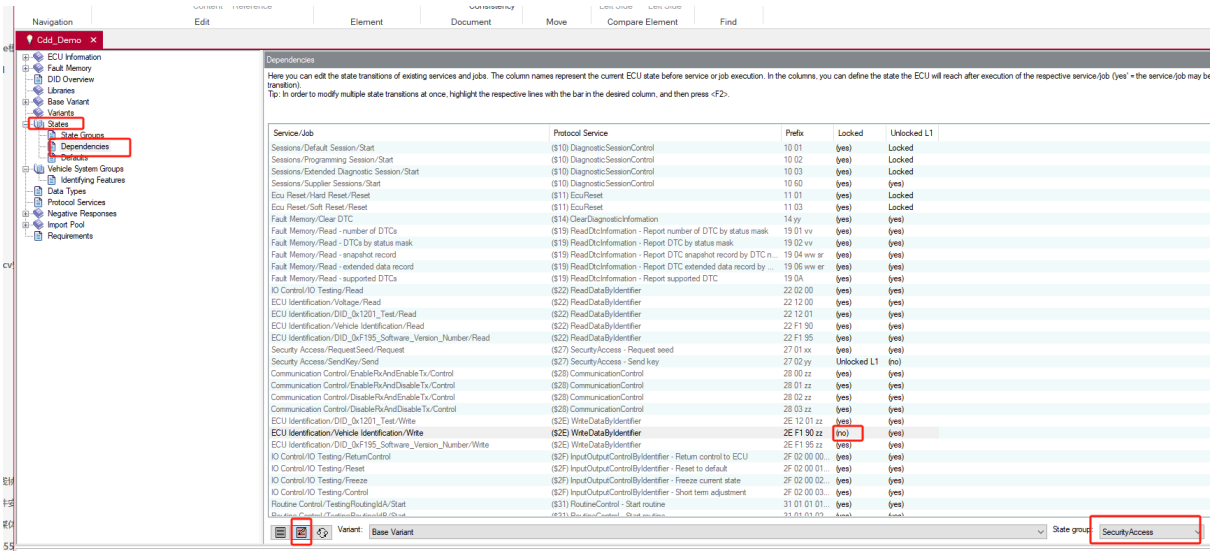
四、VIN码DID 要02会话和安全等级

VIN码DID F195 读没有限制，但是2E写的时候有限制，必须02会话且解锁01, 如下图

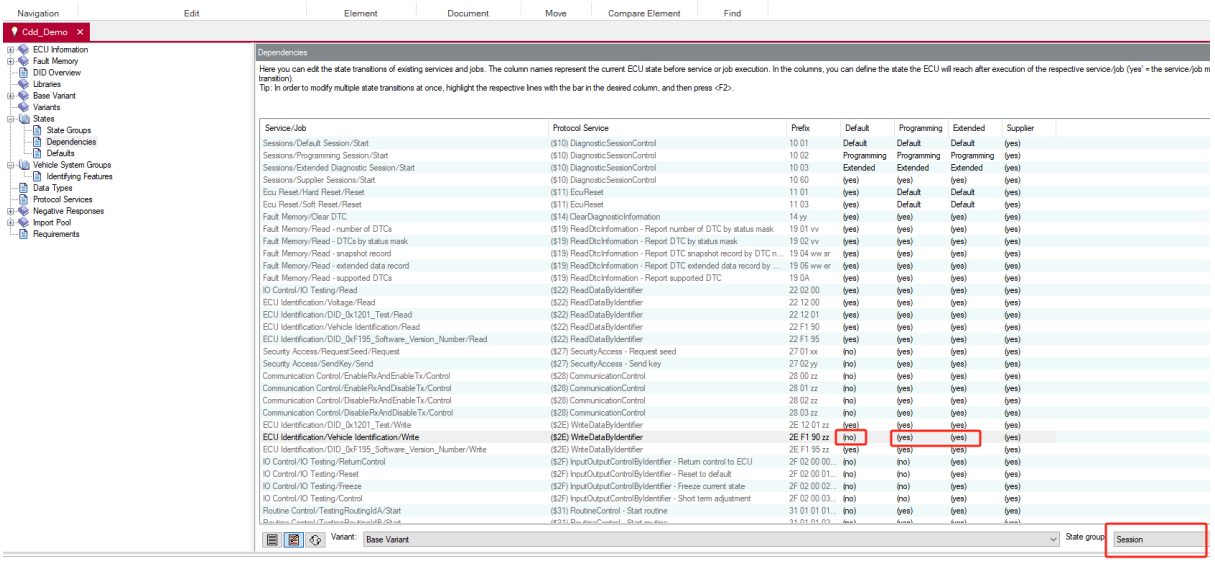


配置来源CDD

其中Locked那一列需要选no，即在未解锁时不可以进行VIN码 2E



会话Default是 no的

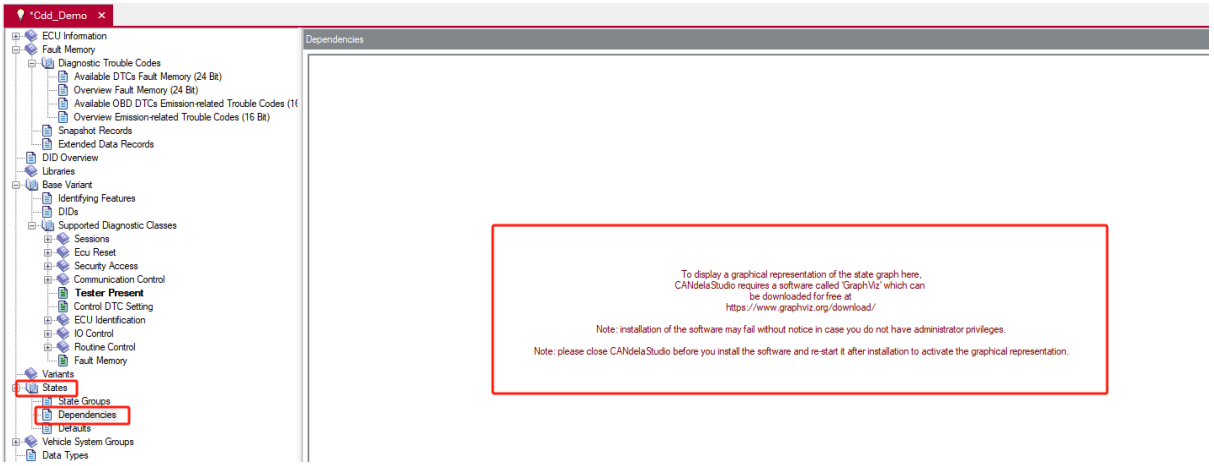


4.1、Depeddenceise 依赖

https://www.graphviz.org/download

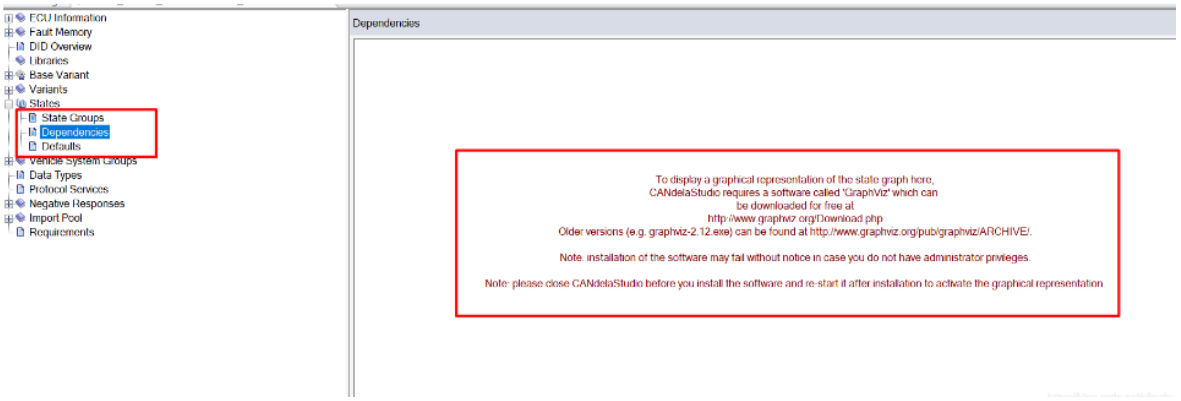
Windows

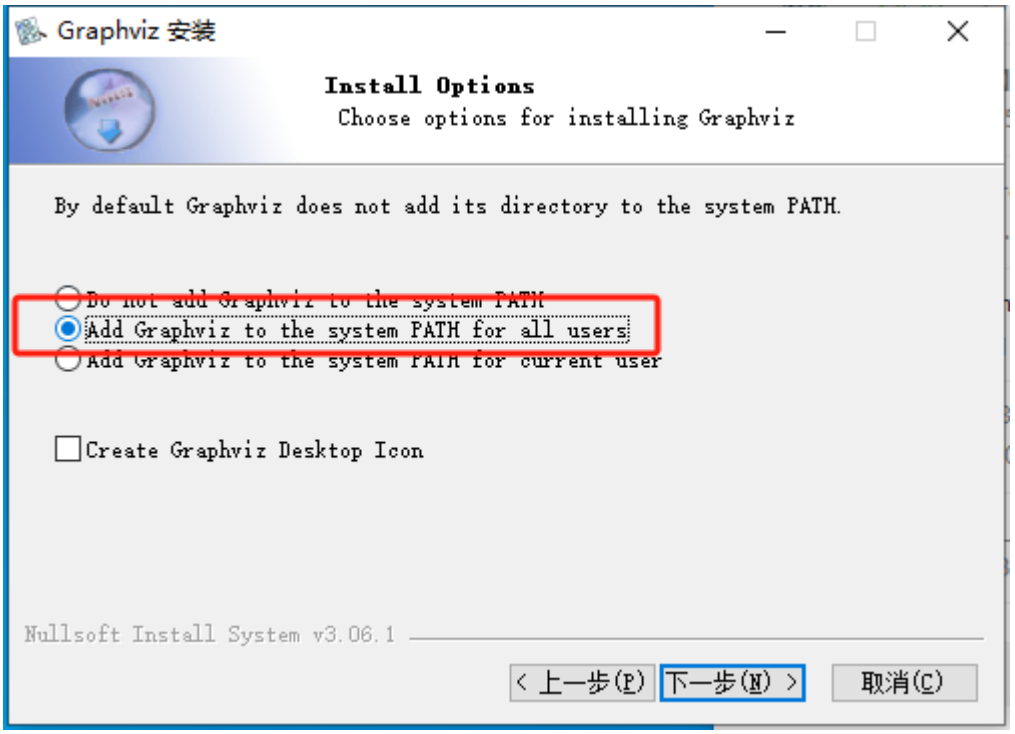
- Stable Windows install packages, built with Microsoft Visual Studio 16 2019:
  - graphviz-13.1.2
    - graphviz-13.1.2 (64-bit) EXE installer [sha256]
    - graphviz-13.1.2 (32-bit) ZIP archive [sha256] (contains all tools and libraries)
    - graphviz-13.1.2 (64-bit) ZIP archive [sha256] (contains all tools and libraries)
    - graphviz-13.1.2 (32-bit) EXE installer [sha256]
  - graphviz-13.1.1
    - graphviz-13.1.1 (64-bit) ZIP archive [sha256] (contains all tools and libraries)
    - graphviz-13.1.1 (64-bit) EXE installer [sha256]
    - graphviz-13.1.1 (32-bit) EXE installer [sha256]
    - graphviz-13.1.1 (32-bit) ZIP archive [sha256] (contains all tools and libraries)
  - graphviz-13.1.0
    - graphviz-13.1.0 (64-bit) EXE installer [sha256]
    - graphviz-13.1.0 (32-bit) ZIP archive [sha256] (contains all tools and libraries)
    - graphviz-13.1.0 (32-bit) EXE installer [sha256]

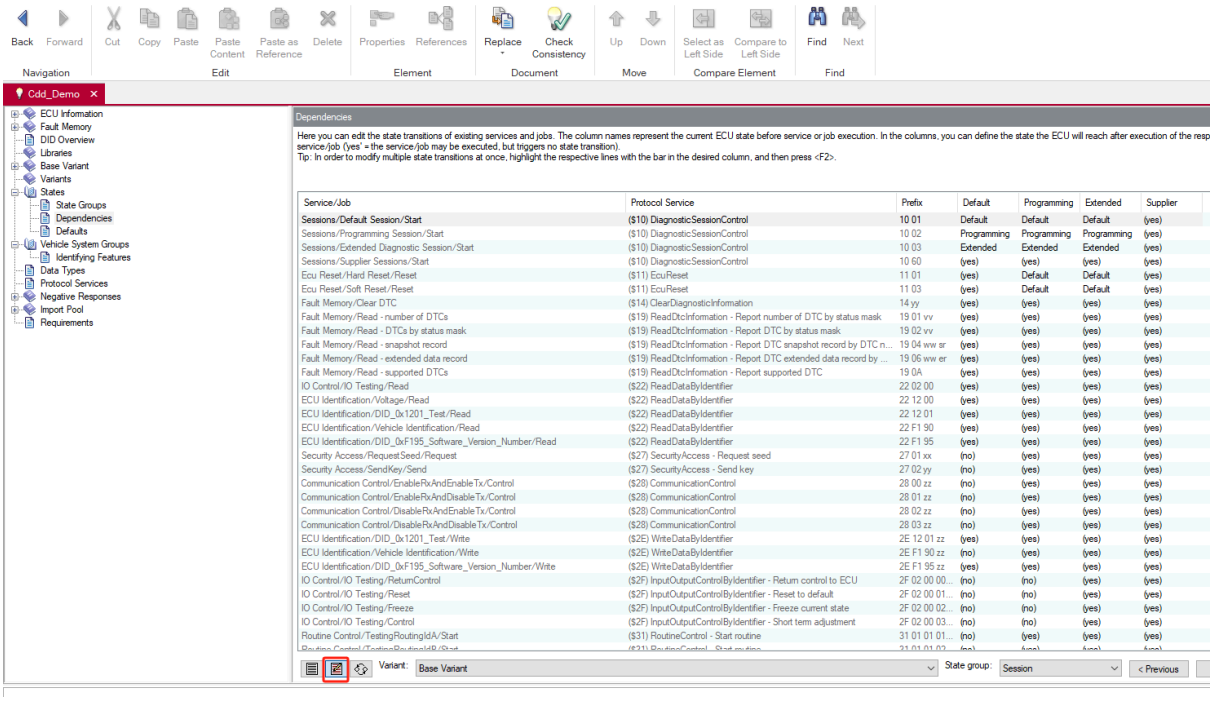


初次用**CANdela**查看“**会话状态机**”和“**安全访问状态机**”（路径：State-->Dependencies）会提示如下错误：

To display a graphical representation of the state graph here,CANdelaStudio requires a software called 'GraphViz' which can be downloaded for free at:http://www. graphviz org/Download.php
Older versions (e.g. graphviz-2.12 exe) can be found at:http://www. graphvz orgpub/graphvz/ARCHIVE/.
Note: installation of the software may fail without notice in case you do not have administrator privileges.Note: please close CANdelaStudio before you install the software and re-start it after installation to activate the graphical representation.






Back
Forward
Cut
Copy
Paste
Paste Content
Paste as Reference
Delete
Properties
References
Replace
Check Consistency
Up
Down
Select as Left Side
Compare to Left Side
Find
Next
Navigation
Edit
Element
Document
Move
Compare Element
Find
Cdd_Demo
ECU Information
Fault Memory
DID Overview
Libraries
Base Variant
Variants
States
State Groups
Dependencies
Defaults
Vehicle System Groups
Identifying Features
Data Types
Protocol Services
Negative Responses
Import Pool
Requirements
Dependencies
Here you can edit the state transitions of existing services and jobs. The column names represent the current ECU state before service or job execution. In the columns, you can define the state the ECU will reach after execution of the resp
service/job (yes' = the service/job may be executed, but triggers no state transition).
Tip: In order to modify multiple state transitions at once, highlight the respective lines with the bar in the desired column, and then press <F2>.
Service/Job | Protocol Service | Prefix | Default | Programming | Extended | Supplier
Sessions/Default Session/Start | ($10) DiagnosticSessionControl | 10 01 | Default | Default | Default | (yes)
Sessions/Programming Session/Start | ($10) DiagnosticSessionControl | 10 02 | Programming | Programming | Programming | (yes)
Sessions/Extended Diagnostic Session/Start | ($10) DiagnosticSessionControl | 10 03 | Extended | Extended | Extended | (yes)
Sessions/Supplier Sessions/Start | ($10) DiagnosticSessionControl | 10 60 | (yes) | (yes) | (yes) | (yes)
Ecu Reset/Hard Reset/Reset | ($11) EcuReset | 11 01 | (yes) | Default | Default | (yes)
Ecu Reset/Soft Reset/Reset | ($11) EcuReset | 11 03 | (yes) | Default | Default | (yes)
Fault Memory/Clear DTC | ($14) ClearDiagnosticInformation | 14 yy | (yes) | (yes) | (yes) | (yes)
Fault Memory/Read - number of DTCs | ($19) ReadDtcInformation - Report number of DTC by status mask | 19 01 vv | (yes) | (yes) | (yes) | (yes)
Fault Memory/Read - DTCs by status mask | ($19) ReadDtcInformation - Report DTC by status mask | 19 02 vv | (yes) | (yes) | (yes) | (yes)
Fault Memory/Read - snapshot record | ($19) ReadDtcInformation - Report DTC snapshot record by DTC n... | 19 04 ww sr | (yes) | (yes) | (yes) | (yes)
Fault Memory/Read - extended data record | ($19) ReadDtcInformation - Report DTC extended data record by ... | 19 06 ww er | (yes) | (yes) | (yes) | (yes)
Fault Memory/Read - supported DTCs | ($19) ReadDtcInformation - Report supported DTC | 19 0A | (yes) | (yes) | (yes) | (yes)
IO Control/IO Testing/Read | ($22) ReadDataByIdentifier | 22 02 00 | (yes) | (yes) | (yes) | (yes)
ECU Identification/Voltage/Read | ($22) ReadDataByIdentifier | 22 12 00 | (yes) | (yes) | (yes) | (yes)
ECU Identification/DID_0x1201_Test/Read | ($22) ReadDataByIdentifier | 22 12 01 | (yes) | (yes) | (yes) | (yes)
ECU Identification/Vehicle Identification/Read | ($22) ReadDataByIdentifier | 22 F1 90 | (yes) | (yes) | (yes) | (yes)
ECU Identification/DID_0xF195_Software_Version_Number/Read | ($22) ReadDataByIdentifier | 22 F1 95 | (yes) | (yes) | (yes) | (yes)
Security Access/RequestSeed/Request | ($27) SecurityAccess - Request seed | 27 01 xx | (no) | (yes) | (yes) | (yes)
Security Access/SendKey/Send | ($27) SecurityAccess - Send key | 27 02 yy | (no) | (yes) | (yes) | (yes)
Communication Control/EnableRxAndEnableTx/Control | ($28) CommunicationControl | 28 00 zz | (no) | (yes) | (yes) | (yes)
Communication Control/EnableRxAndDisableTx/Control | ($28) CommunicationControl | 28 01 zz | (no) | (yes) | (yes) | (yes)
Communication Control/DisableRxAndEnableTx/Control | ($28) CommunicationControl | 28 02 zz | (no) | (yes) | (yes) | (yes)
Communication Control/DisableRxAndDisableTx/Control | ($28) CommunicationControl | 28 03 zz | (no) | (yes) | (yes) | (yes)
ECU Identification/DID_0x1201_Test/Write | ($2E) WriteDataByIdentifier | 2E 12 01 zz | (yes) | (yes) | (yes) | (yes)
ECU Identification/Vehicle Identification/Write | ($2E) WriteDataByIdentifier | 2E F1 90 zz | (no) | (yes) | (yes) | (yes)
ECU Identification/DID_0xF195_Software_Version_Number/Write | ($2E) WriteDataByIdentifier | 2E F1 95 zz | (yes) | (yes) | (yes) | (yes)
IO Control/IO Testing/ReturnControl | ($2F) InputOutputControlByIdentifier - Return control to ECU | 2F 02 00 00... | (no) | (no) | (yes) | (yes)
IO Control/IO Testing/Reset | ($2F) InputOutputControlByIdentifier - Reset to default | 2F 02 00 01... | (no) | (no) | (yes) | (yes)
IO Control/IO Testing/Freeze | ($2F) InputOutputControlByIdentifier - Freeze current state | 2F 02 00 02... | (no) | (no) | (yes) | (yes)
IO Control/IO Testing/Control | ($2F) InputOutputControlByIdentifier - Short term adjustment | 2F 02 00 03... | (no) | (no) | (yes) | (yes)
Routine Control/TestingRoutingIdA/Start | ($31) RoutineControl - Start routine | 31 01 01 01... | (no) | (yes) | (yes) | (yes)
Variant:
Base Variant
State group:
Session
< Previous